Number Systems Code-Lab: Binary and Hexadecimal Conversion

Prof. Jordan C. Hanson

1 Code Lab: `int(x,base)`

1. Python can convert a string representation of a number into decimal using

   ```
   int(x, base)
   ```

   Try:

   ```
   print(int("1010", 2))
   print(int("1111", 2))
   print(int("100000", 2))
   ```

   Record the decimal results:

2. Now try hexadecimal input:

   ```
   print(int("A", 16))
   print(int("1F", 16))
   print(int("2A", 16))
   ```

   Record the decimal results:

2 Code Lab: `bin()` and `hex()`

1. Python converts a decimal integer to binary using `bin()`. Try:

   ```
   print(bin(10))
   print(bin(15))
   print(bin(32))
   ```

   Python includes the prefix `0b` to indicate a binary value.
2. Decimal integers can be converted to hexadecimal using `hex()`. Try:

   ```
   print(hex(10))
   print(hex(31))
   print(hex(42))
   ```

   Python includes the prefix `0x` to indicate hexadecimal.

3 Code Lab: Mixed Practice

1. Before running the code, predict each result:

   ```
   x = int("1101", 2)

   print(x)
   print(bin(x))
   print(hex(x))
   ```

   Decimal:
   Binary:
   Hexadecimal:

2. Try the same sequence with hexadecimal input:

   ```
   x = int("3C", 16)

   print(x)
   print(bin(x))
   print(hex(x))
   ```

   Decimal:
   Binary:
   Hexadecimal:

4 Challenge

1. Use Python to complete the table.

Decimal	Binary	Hex
18		
	101101	
		7F

   Use only

   ```
   int(x, base)
   bin()
   hex()
   ```

   to check your answers.